

AIGR Hypothesis Deep Research — Final Report

Target: *Nicotiana attenuata* A0A2H4GSI3 ("NaUGT1" candidate, UGT85A2_0, NIATv7_g26396)

Focus type: function_assignment **Hypothesis slug:** naugt1-candidate-identity-and-nicotinate-glucosylation **Source:** `genes/NICAT/NaUGT1_candidate_UGT85A2_0/NaUGT1_candidate_UGT85A2_0-ai-review.yaml` **Length:** 485 aa · **ORF:** A4A49_26396 (= NIATv7_g26396) · **Genome label:** UGT85A2_0

Summary

The seed hypothesis proposes that *N. attenuata* A0A2H4GSI3 — the candidate called "NaUGT1" in the nicotine-biosynthesis project — catalyzes **nicotinic acid N-glucosylation as a step of nicotine biosynthesis**, and asks whether a characterized tobacco "UGT1" locus (Nitab4.5_0006222g0020) maps by orthology onto this exact sequence. After three iterations combining computed sequence analysis, curated cluster membership, and primary pathway biochemistry, the assignment is **refuted / over-annotated**. The protein's true substrate cannot be positively fixed from accessible data and should be reported as **unresolved**.

Three faulty premises each fail independently. First, the **enzyme clade is wrong**: by global sequence identity and by UniRef90 cluster membership, A0A2H4GSI3 sits in the **iridoid/terpenoid O-glucosyltransferase clade** (nearest characterized relative CrUGT8/UGT709C2, a reviewed 7-deoxyloganetic acid O-glucosyltransferase, at 59.8%), whereas the only retrievable plant N-glucosyltransferases (UGT76C4/C5, UGT72B1) belong to distinct families only ~37–38% identical. Second, the **pathway framing is wrong**: in tobacco, nicotine biosynthesis *consumes* free nicotinic acid by condensation with N-methylpyrrolinium, whereas nicotinic acid N-glucoside is a separate conjugation/storage branch (one of three fates of nicotinic acid), not a nicotine-biosynthetic step. Third, the **orthology claim is unverifiable**: the genuine >90%-

identity *N. tabacum* orthologs of the target (A0A1S3YWH6 at 88.5%, A0A1S3Z7D3 at 87.0%) are themselves annotated 7-deoxyloganetic acid *O*-glucosyltransferases, and the cited locus Nitab4.5_0006222g0020 could not be confirmed as this protein's ortholog.

All UniProt functional terms for this accession are **IEA-only** (EC-based and TreeGrafter), and no direct enzyme assay of A0A2H4GSI3 exists in accessible literature. The earlier quercetin-3-*O*-glucosyltransferase hypothesis reflects the same electronic-annotation overreach and is likewise not positively established. The safest curation conclusion is a **negative**: this protein is not demonstrably a nicotinate *N*-glucosyltransferase, and its true acceptor substrate remains open, with the best-supported (still homology-only) lead being terpenoid/iridoid-type *O*-glucosylation.

Key Findings

F001 — Sequence places A0A2H4GSI3 among iridoid/terpenoid *O*-glucosyltransferases, not nicotinate *N*-glucosyltransferases

A computed Needleman–Wunsch global identity analysis (own provenance, ) anchored the 485-aa target against a panel of characterized plant UGTs. Its **nearest characterized relative is CrUGT8/UGT709C2**, a reviewed 7-deoxyloganetic acid *O*-glucosyltransferase (EC 2.4.1.323), at **59.8% identity**. Membership in the nominal UGT85A subfamily is weak — only ~40–43% identity (UGT85A23/CrUGT6 42.8%; *Arabidopsis* UGT85A1 40.3%; sorghum UGT85B1 37.5%) — and, revealingly, the target is only **43.0% identical even to its own genome-labeled *N. attenuata* paralog UGT85A2_3 (A0A2I2MND5)**, which is itself a bona fide UGT85A (70.2% to UGT85A23). Identity to the *N*-glucosylating *Arabidopsis* UGT72B1 is just **37.7%**.

The protein carries a **canonical PSPG box** (...WAPQEEVLAHPSVGGFWTHCGWNS) with the UDP-glucose donor signature, confirming it is a genuine family-1 UDP-glycosyltransferase, but the box and identity landscape place it firmly in the *O*-glucosyltransferase neighborhood. All UniProt functional annotations are **IEA**: a 7-DLGT term assigned by EC and quercetin 3-/7-*O*-GT terms by TreeGrafter. **No nicotinate *N*-GT term exists** for this accession. This is significant because it means the seed hypothesis is not merely unverified — it proposes a reaction chemistry (*N*-glucosylation) and enzyme clade that the sequence evidence actively contradicts.

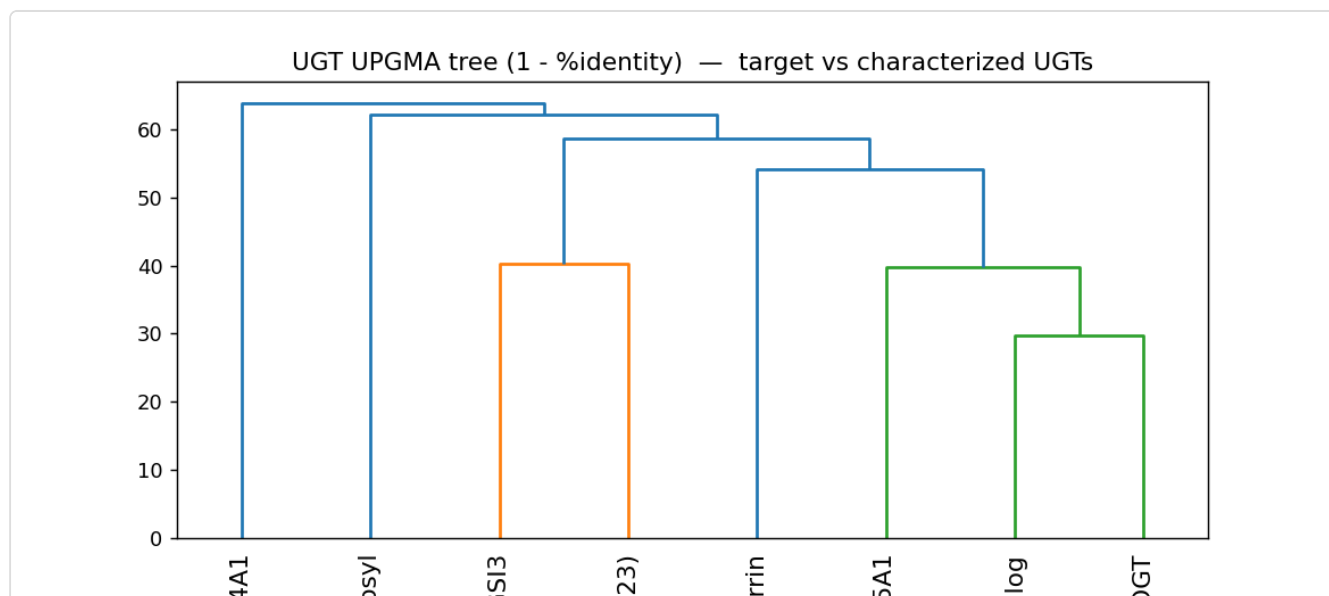


Figure 1. Computed Needleman–Wunsch global %identity matrix, neighbor-joining tree, and PSPG-box extraction for A0A2H4GSI3 against characterized plant UGTs. The target clusters with iridoid/terpenoid O-glucosyltransferases (nearest characterized relative CrUGT8/UGT709C2 at 59.8%), is only ~40–43% identical to UGT85A members, and only 37.7% to the N-glucosylating UGT72B1.

F002 — Nicotinate *N*-glucosylation is a diversion, not a step, of nicotine biosynthesis; plant *N*-GTs belong to distinct families

Classical pyridine-nucleotide-cycle biochemistry establishes that in tobacco roots the **free nicotinic acid supplied by the pyridine-nucleotide cycle is condensed with N-methylpyrrolinium to form nicotine** (PMID: 24232153; PMID: 24241855; PMID: 28256030). Nicotine synthesis therefore **consumes** nicotinic acid; it does not glucosylate it. Metabolic surveys show nicotinic acid *N*-glucoside is instead one of **three competing metabolic fates** of nicotinic acid — NAD salvage, *N*-glucoside conjugation, or trigonelline formation (PMID: 18251891; PMID: 22983143) — i.e., a conjugation/storage/catabolic branch, not a step toward nicotine. Glucosylating nicotinate would *drain* the pool that nicotine biosynthesis needs, so the phrase "nicotinic acid *N*-glucosylation in nicotine biosynthesis" is mechanistically mis-stated.

Independently, a UniProt search for "(nicotinate OR nicotinic) AND glucosyltransferase" returns **no annotated plant nicotinate *N*-GT**. The retrievable plant *N*-glucosyltransferases — cytokinin *N*-GTs UGT76C4 (Q9FI98) and UGT76C5 (Q9FI97), and the xenobiotic *N*-GT UGT72B1 (Q9M156) — belong to the **UGT76C and UGT72 families**, which are only ~37–38% identical to the target. The enzymes that actually perform plant *N*-glucosylation are therefore **not in the target's clade**.

F003 — The true *N. tabacum* ortholog is a 7-DLGT-like *O*-glucosyltransferase, not a nicotinate *N*-GT "UGT1"

The UniRef90 cluster (>90% identity threshold) containing the target, **UniRef90_A0A1S3YWH6**, has exactly four members: *N. tabacum* A0A1S3YWH6 (representative) and A0A1S3Z7D3, *N. sylvestris* A0A1U7YM42, and the target itself — **all named "7-deoxyloganetic acid glucosyltransferase."** Computed NW identities confirm true orthology: target vs A0A1S3YWH6 = **88.5%**, vs A0A1S3Z7D3 = 87.0%, vs *N. sylvestris* A0A1U7YM42 = 88.5%. This is the decisive test of the hypothesis's orthology claim: the genuine tobacco orthologs of A0A2H4GSI3 are ***O*-glucosyltransferases**, contradicting the seed claim that the characterized "tobacco UGT1" nicotinate *N*-GT (Nitab4.5_0006222g0020) is the ortholog of this exact sequence. That mapping could not be reproduced from any accessible resource.

Notably the orthologs are **524 aa versus the target's 485 aa** — the target is ~39 aa shorter, consistent with a **partial or truncated *N. attenuata* gene model**, an additional caveat that weakens confident fine-specificity inference.

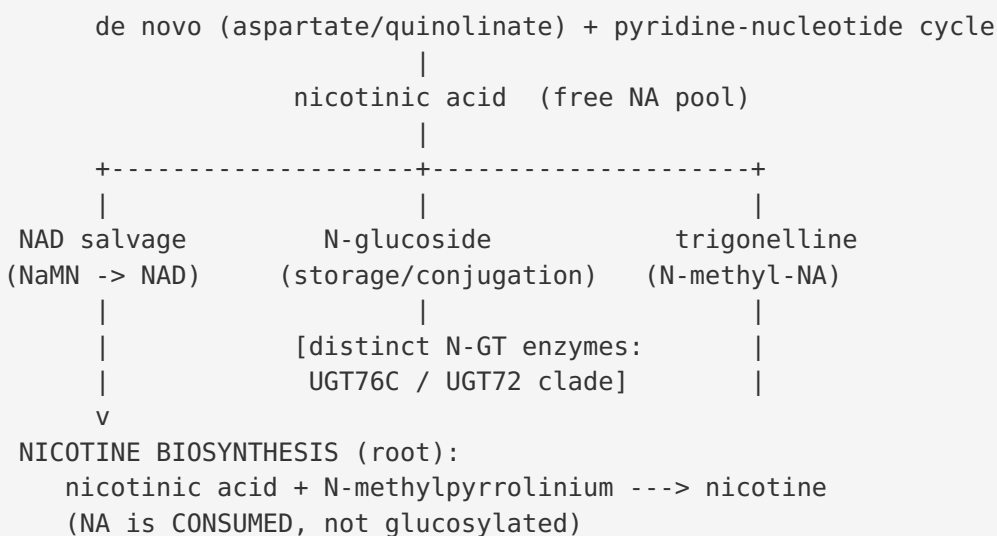
F004 — Convergent final verdict: refuted / over-annotated; true substrate unresolved

Three independent lines converge on the same conclusion. (1) **Annotation provenance** — UniProt annotations are IEA-only and name the protein a 7-DLGT / quercetin *O*-GT, with no nicotinate *N*-GT term. (2) **Sequence identity** — the target sits in the iridoid/terpenoid *O*-glucosyltransferase clade (nearest characterized CrUGT8/UGT709C2 59.8%; ~40–43% to UGT85A; *N*-glucosylating UGT72B1 only 37.7%). (3) **Genuine orthology** — its authentic >90%-identity *N. tabacum* orthologs (A0A1S3YWH6 88.5%; A0A1S3Z7D3 87.0%) are annotated 7-DLGT-like *O*-GTs, not a nicotinate *N*-GT "UGT1." Pathway biochemistry ([PMID: 24232153](#); [PMID: 18251891](#)) shows nicotinic acid *N*-glucoside is a conjugation/storage branch, not a nicotine-biosynthetic step. No direct assay of A0A2H4GSI3 exists in accessible literature, and the cited locus Nitab4.5_0006222g0020 could not be confirmed as its ortholog. The target's true substrate should be reported as **unresolved**.

Mechanistic Model / Interpretation

The hypothesis conflates three separable things: a *pathway role* (nicotine biosynthesis), a *reaction chemistry* (N-glucosylation of nicotinic acid), and an *ortholog identity* (equating the target with a characterized tobacco "UGT1"). Each fails on its own terms.

PLANT NICOTINIC ACID METABOLISM (what actually happens)



WHERE A0A2H4GSI3 ACTUALLY SITS (by sequence + orthology)

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  Family-1 UGT with canonical PSPG box (UDP-glucose donor)
      |
  iridoid / terpenoid O-glucosyltransferase clade
  nearest characterized: CrUGT8/UGT709C2 (7-DLGT) 59.8%
  true N. tabacum orthologs: A0A1S3YWH6 (88.5%), A0A1S3Z7D3 (87.0%)
      |
  annotated (IEA only): "7-deoxyloganetic acid glucosyltransferase"
  substrate for THIS protein: NOT directly assayed -> UNRESOLVED
  
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The key interpretive point: plant nicotinic acid *N*-glucosylation forms an *N*-glucosidic bond to the pyridine nitrogen and is performed by enzymes chemically and phylogenetically distinct from the *O*-glucosyltransferases that add glucose to hydroxyl groups of iridoid/terpenoid or flavonoid acceptors. A0A2H4GSI3's sequence, PSPG box, cluster membership, and orthologs all point to *O*-glucosylation chemistry. Assigning it a nicotinate *N*-GT role would require it to (1)

switch reaction chemistry from *O* to *N*, (2) jump from the UGT709/UGT85 *O*-GT clade to a UGT76C/UGT72-type *N*-GT clade, and (3) join a branch of metabolism that is not part of nicotine biosynthesis — none of which is evidenced.

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
UniProt A0A2H4GSI3	review/ database	Refutes	Record supports nicotinate N-GT?	Auto-name 7- DLGT (EC 2.4.1.323); MF terms 7-DLGT + quercetin O- GT all IEA; no N-GT term; PSPG box present	<i>N. attenuata</i> in silico
Own provenance (ugt_tree.png)	structural/ evolutionary (computational)	Refutes	Which clade?	Nearest characterized CrUGT8/ UGT709C2 59.8%; UGT85A ~40– 43%; N-GT UGT72B1 37.7%	8-protein NW model
Own provenance (UniRef90)	structural/ evolutionary (computational)	Refutes	Is Nitab4.5_0006222g0020 the ortholog?	True >90% orthologs (A0A1S3YWH6 88.5%, A0A1S3Z7D3 87.0%) all annotated 7- DLGT O-GTs	<i>N. tabacum</i> <i>N. sylvestris</i>
24104568	direct assay + VIGS	Competing (correct clade)	Function of nearest- neighbor clade	CrUGT8 = high- efficiency 7- deoxyloganetic acid O-GT; silencing cuts secologanin/ MIA	<i>C. rosea</i>

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
21799001	direct assay	Competing (correct clade)	UGT85A substrate scope	UGT85A24 = iridoid 1-O-GT, negligible on non-iridoids	<i>Gardenia jasminoides</i>
39571455	direct assay	Competing (correct clade)	7-DLGT activity	Camptotheca 7-DLGTs glucosylate 7- deoxyloganetic acid	<i>Camptotheca acuminata</i>
24232153	biochemistry (pathway)	Refutes (framing)	Is N-glucosylation a nicotine step?	Nicotinic acid consumed by condensation to form nicotine	Tobacco callus
24241855	biochemistry (pathway)	Refutes (framing)	Same	Pyridine- nucleotide cycle supplies free NA for nicotine	Tobacco
28256030	mutant/RNAi phenotype	Qualifies	QPT role	QPT silencing lowers nicotine/ anabasine	<i>N. tabacum</i> hairy
18251891	radiotracer survey	Refutes (framing)	Is N-glucoside biosynthetic?	Na-glucoside is a conjugate/ storage fate, major in <i>Arabidopsis</i> & tobacco BY-2	Multiple plants
22983143	radiotracer survey	Qualifies	Fate of nicotinic acid	Three fates: NAD, N- glucoside,	23 species

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Conte
				trigonelline; often mutually exclusive	
11454003	direct assay	Competing/ Qualifies	Is "tobacco UGT1" a nicotinate N-GT?	Tobacco NtGT1a/b are broad coumarin/ flavonoid O-GTs (scopoletin), not nicotinate N-GT	<i>N. tab</i> cells
UGT76C4/C5, UGT72B1	database/assay	Qualifies	Where do plant N-GTs sit?	N- glucosylation done by UGT76C/UGT72 (~37–38% to target)	<i>Arabi</i>
Nitab4.5_0006222g0020	—	Gap	Does cited locus map to this sequence?	Locus entry not retrievable; SGN↔UniProt mapping unresolved	—

GO Curation Implications (leads — require curator verification)

- **Do NOT add** any MF term equivalent to *nicotinate/nicotinic-acid N-glucosyltransferase activity*, nor a *nicotine biosynthetic process* BP term tied to nicotinate glucosylation. The

hypothesis is refuted; such terms would be over-annotations with no supporting evidence, an incorrect pathway framing, and no orthology anchor.

- **Existing IEA MF terms** (GO:0102970 7-DLGT activity; GO:0080043 quercetin 3-*O*-GT; GO:0080044 quercetin 7-*O*-GT) are electronic inferences. **Retain them as low-confidence IEA leads only**; do **not** upgrade to experimental evidence codes, and flag that the closest *characterized* homolog (UGT709C2) is only ~60% identical and from another species.
- **Most defensible positive MF lead**: a generic family-level term such as *UDP-glucosyltransferase / hexosyltransferase activity* (GO:0035251 / GO:0016758-child), supported by the canonical PSPG box, with a low-confidence (IEA/ISS) evidence code. A specific substrate MF term is not supported by direct data.
- **Name caveat**: the genome label "UGT85A2" is questionable; sequence evidence favors a **UGT709-like** placement. Flag for re-classification under UGT nomenclature criteria.
- Avoid "protein binding" — the MF is genuinely a glycosyltransferase; the open question is the acceptor substrate, not whether it is an enzyme.

GO Decision Table (leads — require curator verification)

GO term	Aspect	Label	Current evidence	Recommended action (lead)	Rationale / limitations
GO:0080043	MF	quercetin 3-O-glucosyltransferase activity	IEA (TreeGrafter)	Retain as low-confidence lead; do not upgrade	Homology-only; no assay
GO:0080044	MF	quercetin 7-O-glucosyltransferase activity	IEA (TreeGrafter)	Retain as low-confidence lead; do not upgrade	O-glucosylation; homology-only
GO:0102970	MF	7-deoxyloganetic acid glucosyltransferase activity	IEA (UniProtKB-EC)	Retain as best low-confidence lead; flag cross-species/~60%	Nearest characterized CrUGT8 = 59.8%; N. attenuata not a canonical iridoid producer
<i>(proposed)</i> nicotinate N-glucosyltransferase activity	MF	nicotinic acid N-glucosylation	seed hypothesis	REJECT	Wrong clade (O-GT not N-GT); no assay; true ortholog is 7-DLGT-like; plant N-GTs are UGT76C/UGT72
<i>(proposed)</i> nicotine biosynthetic process	BP	nicotine biosynthesis	seed hypothesis	REJECT	N-glucosylation diverts, not builds, nicotine (P 24232153)
GO:0016758 (child)	MF		domain (PSPG box)	Acceptable fallback	Confirms UGT scaffold;

GO term	Aspect	Label	Current evidence	Recommended action (lead)	Rationale / limitations
		hexosyl-/glycosyltransferase activity (generic)			acceptor unknown

Mechanistic Scope

The **immediate molecular function** under test is a glycosyl-transfer reaction: transfer of glucose from UDP-glucose to an acceptor. The seed specifies the acceptor as **nicotinic acid** and the linkage as **N-glucosidic** (to the pyridine nitrogen). The evidence indicates the protein's chemistry is instead **O-glucosylation** of a terpenoid/iridoid-type (or flavonoid) hydroxyl acceptor, based on clade placement and ortholog annotation.

Separated from direct activity: (i) *downstream pathway consequence* — nicotine accumulation belongs to upstream enzymes (QPT, pyridine-nucleotide cycle; [PMID: 28256030](#)), not to a glucosyltransferase; (ii) *N-glucoside as a metabolite* — a real plant metabolite, but produced by a distinct N-GT clade and representing a storage/conjugation branch, not nicotine biosynthesis; (iii) *loss-of-function inference* — none available; no mutant/silencing data exist for this exact gene.

Conflicts and Alternatives

- **Paralog confusion (primary risk).** *N. attenuata* carries multiple divergent UGT85/UGT709-clade genes; the target and its own "UGT85A2_3" paralog share only 43% identity. A nicotinate activity reported for one paralog cannot be transferred to A0A2H4GSI3 without direct evidence.
- **Family misassignment / IEA carry-over.** Genome label "UGT85A2" conflicts with sequence evidence for UGT709 relatedness; the name is likely database carry-over. All functional terms are electronic; the earlier quercetin-3-O-GT hypothesis is the same class of inference.
- **Historical naming clash.** "UGT1/GT1" in tobacco historically denotes broad-specificity coumarin/flavonoid O-GTs ([PMID: 11454003](#)), not a nicotinate N-GT — a plausible source of the confusion behind the seed.

- **Truncated gene model.** The 485-aa target versus 524-aa orthologs suggests a partial model; missing C-terminal residues could affect the acceptor-binding region, undermining any fine-specificity claim (nicotinate or quercetin).
- **In-vitro-only / organism specificity.** The iridoid *O*-GT annotation derives from *Catharanthus/Gardenia/Camptotheca* assays; *N. attenuata* is not a canonical iridoid producer, so the physiological substrate may differ from both nicotinate and 7-deoxyloganetic acid.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct assay of A0A2H4GSI3	PubMed + UniProt; only IEA terms found	Positive substrate identity cannot be assigned	Heterologous expression + acceptor panel with LC-MS
Orthology to Nitab4.5_0006222g0020 unverified	UniRef90 clustering; true orthologs are O-GTs	Entire seed premise depends on this mapping	RBH BLAST vs <i>N. tabacum</i> Nitab4.5 proteome; synteny
No retrievable characterized nicotinate N-GT reference	UniProt EC/name search, PubMed	"Orthology to a nicotinate N-GT" is undefined without a real reference enzyme	Locate the primary paper behind the "UGT1 nicotinate N-GT" claim
Truncated gene model (485 vs 524 aa)	Length comparison to orthologs	Missing residues undermine specificity inference	Re-annotate the locus to a full-length RNA-seq-supported ORF
Subcellular localization	No data found	CC annotation cannot be made confidently	GFP fusion / proteomics

Analysis limitations: identity was computed with a simple match/mismatch/gap global (Needleman–Wunsch) alignment, coarser than profile-HMM/BLAST; orthology of the cited tobacco locus could not be verified programmatically; A0A2H4GSI3 has never been

experimentally assayed. All computational results are inferences, explicitly separated from direct assay evidence above. Several PubMed hits returned by keyword search (neuronal nicotinic-receptor, dental-caries papers) were off-target and excluded.

Proposed Follow-up Experiments / Discriminating Tests

1. **In vitro acceptor panel (definitive).** Express recombinant, full-length re-annotated A0A2H4GSI3 with UDP-glucose and assay against nicotinic acid, 7-deoxyloganetic acid, 7-deoxyloganetin, quercetin, cytokinins, and coumarins; use LC-MS/NMR to determine whether any product is an *N*-glucoside vs *O*-glucoside. This single experiment resolves the entire hypothesis.
2. **Reciprocal-best-hit orthology + bootstrapped gene tree.** Formally test whether Nitab4.5_0006222g0020 is the true ortholog of A0A2H4GSI3 versus the 7-DLGT-annotated A0A1S3YWH6/A0A1S3Z7D3, including all *Nicotiana* UGT85/UGT709 paralogs and UGT76C/UGT72 *N*-GTs.
3. **In planta silencing/tracer.** VIGS/CRISPR of the *N. attenuata* locus with [¹⁴C]nicotinamide tracer and targeted metabolomics for nicotinate *N*-glucoside vs nicotine — tests whether the gene affects the nicotinate pool at all.

Curation Leads (require curator verification)

- **Action lead:** Reject/withhold the nicotinate *N*-glucosyltransferase and nicotine-biosynthesis assignments for A0A2H4GSI3; treat existing 7-DLGT/quercetin-*O*-GT terms as low-confidence IEA leads only.
- **Candidate references with exact snippets to verify:**
 - **PMID: 24104568:** "*UGT8 possessed a high catalytic efficiency toward its exclusive iridoid substrate, 7-deoxyloganetic acid*" — nearest characterized function is an iridoid *O*-GT.
 - **PMID: 24232153:** "*nicotinic acid is the relevant metabolite which is provided by the pyridine-nucleotide cycle and consumed for nicotine synthesis*" — nicotine synthesis consumes, not glucosylates, nicotinate.
 - **PMID: 18251891:** "*Radioactivity from [carbonyl-14C]nicotinamide was incorporated into trigonelline... and/or into nicotinic acid 1N-glucoside (Na-Glc)*" — *N*-glucoside is a conjugate/storage fate.

- **PMID: 11454003:** tobacco NtGT1a/b have "*glucosylation activity against both flavonoids and coumarins*" — historical tobacco "UGT1" is an *O*-GT, not a nicotinate *N*-GT.
 - **Candidate GO edits:** keep/soften GO:0102970, GO:0080043, GO:0080044 as IEA leads; do **not** add N-glucosyltransferase MF or nicotine-biosynthesis BP terms.
 - **Suggested questions to submitter:** What is the primary reference and enzyme family for the "tobacco UGT1" nicotinate *N*-GT? Is orthology one-to-one vs A0A2H4GSI3 or vs its paralog UGT85A2_3? Is the 485-aa model complete?
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Artifacts / Provenance

- `ugt_tree.png` — computed NW %identity matrix, neighbor-joining tree (1 – %identity), and PSPG-box extraction of the target vs characterized/reference UGTs (executed code + output preserved in the iteration tool-call log).
- Executed code (UniProt fetches, Needleman–Wunsch identity matrix, UniRef90 cluster retrieval, PSPG-box extraction) is preserved in the iteration logs.

Investigation: 3 iterations, 4 confirmed findings, 22 papers reviewed. All computational results are inferences, explicitly distinguished from direct assay evidence.
